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Approved by: Head of Safety & Fire Services

PROCEDURE FOR HOT WORK PERMIT AND FIRE SAFETY
IN PROCESS AREAS

1. PURPOSE & SCOPE

This procedure defines the minimum safety requirements for carrying out hot work (welding, cutting, grinding, brazing, and any spark-producing activity) in all process units, tank farms, and loading gantries of Gujarat Refinery. It applies to all employees, contractors, and vendor personnel.

2. REFERENCES (REGULATORY & CORPORATE)

- OISD-117: Standard on Fire Protection Facilities for Petroleum Refineries and Petrochemical Plants (latest edition).
 - OISD-118: Standard on Layout Requirements for Industrial Installations (fire separation distances).
 - OISD-105: Work Permit System.
 - Petroleum Rules 2002, framed under the Petroleum Act 1934.
 - The Factories Act 1948, Chapter IVA (provisions relating to hazardous processes), Section 31 to 40.
 - PESO (Petroleum and Explosives Safety Organisation) guidelines on storage of flammable liquids.
 - NFPA 51B: Standard for Fire Prevention During Welding, Cutting, and Other Hot Work.
 - API RP 2201: Safe Welding, Cutting and Hot Work Practices.
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3. RESPONSIBILITIES

- 3.1 Performing Authority (Hot Work Executor):
- Vikram Singh (Senior Welding Foreman)
 - Obtain a valid hot work permit before commencing any work.
- 3.2 Area Authority (Plant Incharge):
- Rajesh Khanna (Process Manager, FCC Unit)
 - Ensure the worksite is made safe; verify gas-test results.
- 3.3 Safety Officer:
- Mohammad Iqbal (Fire & Safety Officer)
 - Issue the permit only after verifying all controls per OISD-105.
- 3.4 Gas Tester:
- Sunil Yadav (Certified Gas Tester, certificate valid till 12-Dec-2025)

4. PRE-WORK REQUIREMENTS (CONTROLS)

- a) Conduct Gas Free Test (GFT): LEL must be below 0% (zero) within a 10-metre radius of the hot-work spot. Use a calibrated combustible gas detector.
- b) Remove all combustible material within an 11-metre (35 ft) radius, OR cover with fire-resistant blankets (per OISD-117, clause 6.3).
- c) Provide a minimum of two 9-kg Dry Chemical Powder (DCP) fire extinguishers and one 50-litre foam trolley at the worksite.
- d) Position a fire-watch with a charged fire hose throughout the work and for a minimum of 30 minutes after completion.
- e) Earth all welding equipment; ensure continuity per IS 3043.
- f) Display the hot-work permit at the worksite.
- g) Stop all hot work if wind speed exceeds 35 km/h or during a thunderstorm warning.

5. EQUIPMENT & MONITORING

- Combustible gas detector (MSA Altair 5X), calibrated 20-Feb-2024.
- Equipment isolation: E-305 (Kettle reboiler) and V-203 (suction scrubber) drained and blinded; blinds recorded in blind register BR-2024-0042.
- Monitor suction scrubber V-203 pressure; must remain below 0.5 kg/cm²g during hot work in the vicinity.

6. PERMIT VALIDITY & CLOSURE

- A hot work permit is valid for a single shift (max 8 hours) from the issue time.
- Closure requires sign-off by Performing Authority, Area Authority, and Safety Officer, with a final fire-watch confirmation.
- Retain the closed permit for a minimum of 2 years (Factories Act 1948, Section 12 records).

7. NON-COMPLIANCE

Any violation is a reportable safety incident per OISD-105 and shall be logged in the refinery incident management system (IMS) within 24 hours. Repeat violations may attract penalties under the Factories Act 1948 and contract termination.

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